FarmFederate: Federated Vision-Language Learning for Tea Disease Diagnosis

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***Abstract*—Tea disease diagnosis in the field rarely starts from an image alone. Agronomists also record lesion shape, affected shoots, recent weather, and soil conditions, but most automated systems still treat disease recognition as either an image-only or text-only task. Centralizing these records is also awkward for tea estates: high-resolution leaf photographs are bandwidth-heavy, and field logs can be commercially sensitive. We present FarmFederate, a multimodal federated learning framework for five common tea disease classes (*leaf blight, leaf hoppers, leaf rust, looper caterpillars, mosquito bug*). The system pairs lightweight LLM text encoders with ViT-style image encoders and evaluates eight VLM fusion strategies under FedAvg coordination, so only model updates leave the estate device. For explanation, uploaded leaf photographs are marked with class-aware diseased-region masks and oriented bounding boxes (OBBs), while text-only symptom descriptions retrieve representative YOLO-OBB reference images. On a localized corpus containing 200 field photographs, 371 YOLO-OBB annotations, 792 training-run image tensors after minority-class fill, and 3,000 agronomic text samples, the strongest text-only and image-only baselines obtain macro F1 scores of 0.487 and 0.911. The best multimodal configuration, CLIP-style contrastive fusion, reaches 0.949. Across three simulated non-IID garden clients, federated training preserves centralized performance while keeping raw images, notes, and retrieval records local. Against a 35-model tea-disease benchmark, FarmFederate remains in the leading group while adding privacy-preserving multimodal diagnosis and local advisory retrieval.**

***Index Terms*—Federated Learning, Vision-Language Models, Tea Leaf Disease Classification, Multimodal Learning, Precision Agriculture, Privacy-Preserving Machine Learning**

I. Introduction

TEA (*Camellia sinensis*) is a vital cash crop for millions of smallholder farmers across South Asia, East Africa, and East Asia. Five diseases, leaf blight, leaf hoppers, leaf rust, looper caterpillars, and mosquito bug, can cut yields and reduce leaf quality across major growing regions [1], [2]. Recent diagnostic systems have improved field deployment through lightweight CNNs, attention modules, and IoT-aware pipelines [1]–[5], [7]. Most of these systems, however, still begin and end with the photograph. That is not how tea diagnosis is usually recorded on the ground: field staff note lesion shape, flush condition, weather history, soil status, and pest pressure alongside the image. A purely visual classifier discards this second record. At the same time, sending every high-resolution image and field note to a central repository is not attractive for estates with limited connectivity and proprietary cultivation records. Federated Learning (FL) [14] addresses the second problem by keeping data local, but existing FL work on tea pathology has not yet integrated the visual and textual evidence used in real inspections.

A. Motivation

The motivation is therefore operational, not only architectural. In a tea garden, two leaves with similar brown patches may be interpreted differently after recent rainfall, dry wind, or insect activity is considered. Conversely, a symptom note without the corresponding leaf texture can confuse rust pustules, necrotic blight, and pest injury. FarmFederate treats these two sources as complementary rather than competing inputs. The image branch captures colour, shape, and lesion geometry; the text branch carries the agronomic context that is usually written in field logs.

The same design also respects estate-level data ownership. Raw photographs, symptom notes, IoT context, and retrieval documents remain on the local device, while FedAvg exchanges only model updates. For interpretability, the diagnostic visualizer highlights diseased regions in uploaded leaf images and draws OBBs around affected patches. For text-only prompts, it retrieves annotated OBB reference images instead of generating synthetic pictures, keeping the explanation tied to observed disease examples.

B. Contributions

This paper makes five concrete contributions:

- We evaluate tea disease diagnosis as an image-text problem rather than an image-only benchmark. Under the same non-IID FedAvg setup, 18 variants are tested: 5 LLM text encoders, 5 ViT image encoders, and 8 VLM fusion strategies. VLM fusion reaches macro F1 of 0.886–0.949, above the text-only range of 0.417–0.487 and the image-only range of 0.861–0.911.
- We build a localized, ambiguity-aware tea pathology corpus from 200 field-collected photographs, 371 YOLO-OBB disease annotations, and 3 000 class-balanced agronomic symptom records. The reported training run uses augmentation-based minority-class fill to obtain 792 image tensors. The text set deliberately mixes cross-class cues (8% clear, 22% slightly mixed, 70% heavily mixed), so a model cannot succeed by memorising disease words.
- We add an anti-collapse training stack for rare and imbalanced field classes: balanced batch sampling, focal loss [19], entropy-based diversity regularisation, square-root-dampened class weights capped at 10×, and early collapse abort. The stack keeps all five classes active even when the local data split is skewed.
- We quantify the cost of keeping estate data local. Across three simulated non-IID garden clients, federated training retains centralised performance on average: LLM 107.7% (0.460 vs. 0.427), ViT 100.0% (0.886 vs. 0.886), and VLM 98.6% (0.861 vs. 0.873), while raw leaf images, field notes, and knowledge-base documents remain on local devices.
- We connect classification to an advisory workflow through a local Classify–Retrieve–Advise RAG pipeline, per-estate FAISS knowledge bases, IoT-aware re-ranking, and visual explainability. Uploaded images receive class-aware diseased-region masks and OBBs; text-only queries retrieve representative YOLO-OBB reference images with disease tags.

II. Related Work

A. Tea Leaf Disease Detection

Recent tea-specific diagnosis has moved from classical image descriptors toward lightweight and attention-based deep models. Yang *et al.* [1] introduced WaveLiteNet, a wavelet-CNN tea disease model that reported 98.70% accuracy with only 3.16 million parameters. Hu *et al.* [2] combined lightweight tea-leaf segmentation with multiscale group-attention residual classification and achieved 97.43% accuracy on field images. Deng and Photong [3] evaluated VGG16, ResNet50, and DenseNet169 for tea leaf disease recognition in an IEEE-published study. These systems strengthen tea-specific visual diagnosis, but they remain image-only and centrally trained.

Recent IEEE work on plant disease detection also emphasizes practical deployment. Joseph *et al.* [4] released a real-time plant disease dataset and detector in *IEEE Access*. Maurya *et al.* [5] proposed a lightweight meta-ensemble suitable for IoT environments, while PlantDet [6] used a robust multi-model ensemble with Grad-CAM and Score-CAM explanations. Al-Shahari *et al.* [7] integrated IoT-assisted crop monitoring with deep disease detection, and Shafik *et al.* [8] surveyed plant disease datasets, model families, and deployment challenges. These high-quality recent studies improve field-ready plant diagnosis, but they still do not combine farmer text logs with leaf images under federated training.

B. Federated Learning for Tea and Plant Disease Detection

McMahan *et al.* [14] introduced FedAvg, which remains the core aggregation baseline for decentralized model training. More recent agriculture work has moved FL from generic benchmarks toward smart-farm deployment. Tahir *et al.* [9] proposed a federated explainable-AI framework for smart agriculture in *IEEE Access*. Devaraj *et al.* [10] introduced RuralAI, a hierarchical FL architecture for crop-health monitoring in *IEEE Sensors Letters*. Ding *et al.* [11] designed Bc^{2}FL, a double-layer blockchain-driven FL framework for Agricultural IoT in the *IEEE Internet of Things Journal*. Caminha and Oliveira [12] studied cloud-driven FL for plant disease detection, and Thonglek and Rattanatamrong [13] explored decentralized FL for plant disease identification on edge devices.

These works show that FL is becoming practical for distributed farms, edge devices, and Agricultural IoT. However, they use image or sensor streams only. None pairs leaf photographs with agronomist text annotations in a federated tea-disease setting.

C. Vision-Language Models for Agriculture

CLIP [20] showed contrastive vision-language pre-training enables zero-shot transfer. BLIP-2 [21] introduced the Q-Former for vision-language alignment. Flamingo [22] proposed gated cross-attention for few-shot visual language modelling. CoCa [23] unified contrastive and captioning objectives into one model. Wang *et al.* [25] proposed LLMI-CDP, a multimodal large-language model for crop disease and pest identification using LoRA fine-tuning. Li *et al.* [26] proposed FedReplay, which combines frozen CLIP features with federated transfer learning for privacy-preserving smart agriculture. These newer multimodal systems confirm that structured text can complement visual symptoms, but tea-specific multimodal FL remains unaddressed.

FarmFederate builds on this line of work by combining LLM text encoders, ViT image encoders, and eight VLM fusion strategies inside federated training rounds. To the best of our knowledge, this combination has not previously been evaluated for tea leaf disease diagnosis.

TABLE I: Comparison with Related Work on Tea/Plant Disease Detection.

	Modality		Training	
Work	**Image**	**Text**	**FL**	**VLM**
Tea Leaf Disease Detection				
Yang [1]	✓	×	×	×
Hu [2]	✓	×	×	×
Deng [3]	✓	×	×	×
Joseph [4]	✓	×	×	×
Maurya [5]	✓	×	×	×
Federated Learning for Disease				
Tahir [9]	✓	×	✓	×
Devaraj [10]	✓	×	✓	×
Ding [11]	✓	×	✓	×
Caminha [12]	✓	×	✓	×
Multimodal / VLM for Agriculture				
Wang [25]	✓	✓	×	✓
Li [26]	✓	✓	✓	✓
FarmFederate (ours)	✓	✓	✓	✓

Synthesis. The gap is therefore not the absence of strong image classifiers, FL protocols, or VLM components in isolation. Rather, these threads have rarely been tested in the same tea-garden setting, where symptom notes, leaf photographs, privacy constraints, and treatment retrieval all matter at once. Existing tea models mostly optimize centralized visual recognition; agricultural FL studies mostly protect distributed data without multimodal diagnosis; and VLM systems show useful image-text alignment without addressing estate-level data ownership. FarmFederate targets this missing junction by evaluating tea-field images, agronomist text, federated aggregation, VLM fusion, local retrieval, and OBB-based visual explanation in one workflow.

Table I shows that FarmFederate is the only tea-focused framework to combine all four capabilities. It uses image and text modalities, federated learning, and VLM fusion, and is the first to apply this combination to tea leaf disease detection.

III. System Model

A. Architecture Overview

Figure 1 shows the FarmFederate architecture. Each tea estate holds three types of data. Leaf photographs come from drones or smartphones. Agronomists write field annotations describing lesion shape and growing conditions. IoT sensors record temperature, humidity, soil moisture, pH, and EC. All of this stays on the local device. Only model weights travel to the aggregation server.

Data moves through three stages on each garden device:

1) **Preprocessing:** Field annotations are tokenised with WordPiece [17] (up to 128 tokens); leaf photographs are resized to 224×224 and normalised with standard ImageNet channel statistics.
2) **Encoding:** An LLM encoder reads the annotation text and produces a feature vector $\mathbf{h}_t \in \mathbb{R}^{256}$ from the `[CLS]` token. A ViT encoder reads the leaf image and produces $\mathbf{h}_v \in \mathbb{R}^{512}$ via adaptive average pooling.
3) **Fusion and output:** A VLM fusion module combines $\mathbf{h}_t$ and $\mathbf{h}_v$ into a joint representation $\mathbf{h}_f$, which a classification head maps to one of the five tea leaf diseases.

B. Dataset Description

We evaluate FarmFederate on a fully localized image dataset leveraging original field collections from the Tea Garden of the Indian Institute of Technology Kharagpur (IIT Kharagpur). The dataset is paired with a class-balanced synthetic agronomic text corpus. The five disease classes

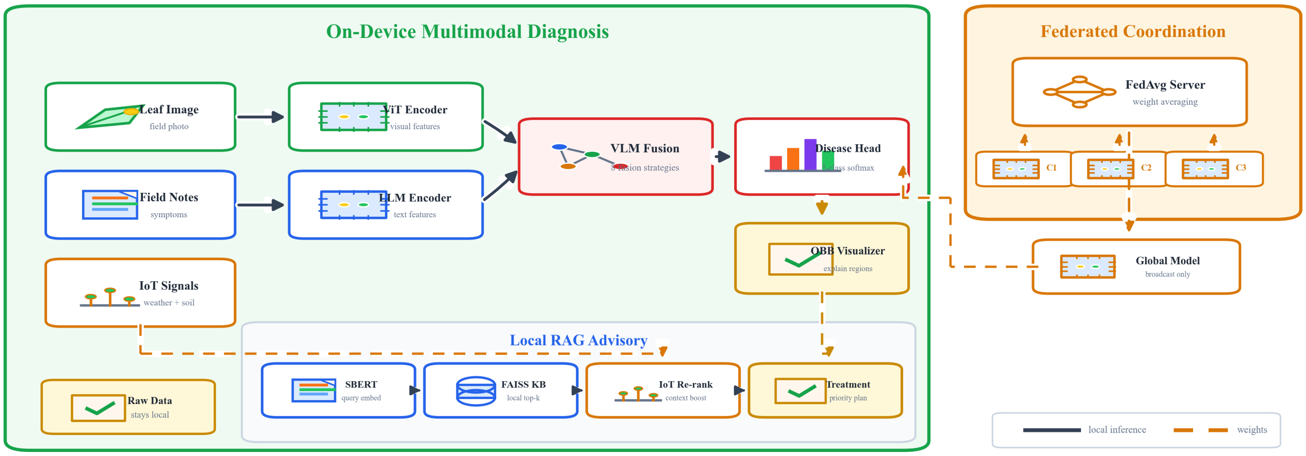


Fig. 1: Illustrative FarmFederate workflow: local tea-garden devices fuse leaf images, field notes, and IoT context for disease diagnosis, OBB-based explanation, and RAG advisory; FedAvg coordinates only model-weight updates.

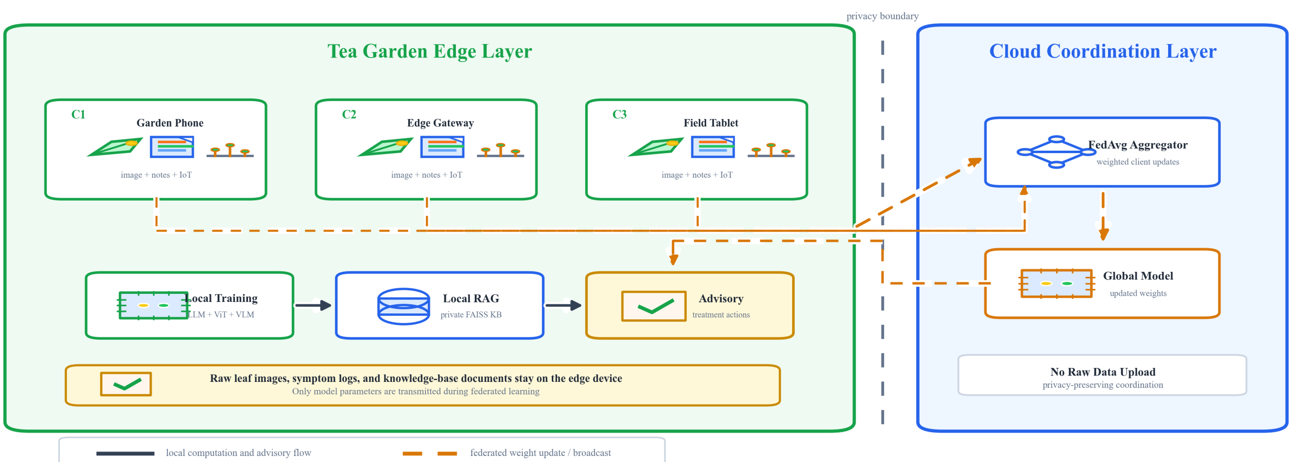
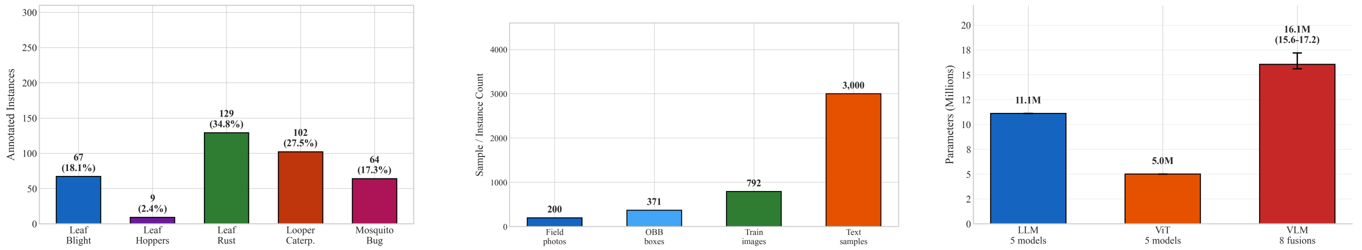


Fig. 2: Edge-cloud compute allocation and privacy boundary. Tea-garden clients perform local multimodal training, classification, FAISS retrieval, and advisory generation; the cloud server aggregates FedAvg weight updates and broadcasts the global model without receiving raw farm data.

are: *leaf_blight* (fungal infection causing brown water-soaked necrosis), *leaf_hoppers* (insect-mediated puncture lesions), *leaf_rust* (orange-yellow rust pustules on the lower leaf surface), *looper_caterpillars* (ragged feeding holes and skeletonised patches), and *mosquito_bug* (raised corky feeding wounds and shoot-tip dieback). Together, these classes cover fungal spotting, rust pustules, and insect feeding damage, requiring the model to separate both texture cues and pest-injury patterns.

1) Image Data: The visual corpus originates exclusively from field collections. **Field-collected:** 200 high-resolution photographs captured during active growing seasons on-site at the IIT Kharagpur tea plantation. The sorted real-image folders contain 35 leaf-blight, 8 leaf-hoppers, 62 leaf-rust, 48 looper-caterpillars, and 47 mosquito-bug images. Annotations strictly use YOLO Oriented Bounding Box (OBB) format to handle non-axis-aligned disease patches, yielding 371 annotated disease boxes. The Colab setup also prepares a balanced export folder with 800 images per class for inspection and download, but the reported training run loads the sorted real dataset and applies synthetic fill only to the under-represented leaf-hoppers class. This yields 792 image tensors in the training run (200 real images plus 592 generated minority-fill images), split into 633 train and 79 validation images. All images are resized to 224×224 and normalised with ImageNet statistics.

Here, *synthetic fill* means augmentation-derived minority sampling rather than text-to-image generation. No diffusion model, GAN, or external image generator is used for the reported result. The scarce leaf-hoppers photographs are repeatedly sampled and transformed using label-preserving operations such as horizontal/vertical flips, right-angle and small-angle rotations, brightness/contrast/saturation jitter, autocontrast, sharpening, and additive noise. These augmented tensors increase exposure to the rare class while preserving the field-



(a) Tea disease class distribution in the field-collected YOLO-OBB annotations.

(b) Dataset: 200 field images, 371 OBB boxes, 792 training-run image tensors, and 3 000 text samples.

(c) Parameter ranges across 18 architectures. VLM fusions heaviest; ViT lightest.

Fig. 3: Dataset statistics and model sizes: disease-class distribution (left), field-data augmentation summary (centre), and parameter ranges (right).

TABLE II: Dataset Overview

Category	Parameter	Value
General	Num. classes	5
	Text source	Controlled agronomic templates
	Image source	IIT Kharagpur Tea Garden
	Train/Val/Test	Stratified, run-specific
Image	Base raw images	200 (YOLO-OBB annotated)
	OBB instances	371 annotated boxes
	Training-run images	792 (200 real + 592 fill)
	Fill method	Augmentation only; no GAN/diffusion
	Image Train / Val	633 / 79
	Normalisation	ImageNet mean/std
	Resolution	224×224
Text	Total samples	3 000 (600/class)
	Generator	Template slots + symptom vocabularies
	Diversity split	8% clear, 22% slight, 70% heavy
	Max seq. length	128 tokens
	Train / Val	2 400 / 300

image origin of the visual corpus. Procedural class-pattern images can be created by the data-preparation code as a fallback for empty classes or visual demonstrations, but they are not the basis of the reported 200-real plus 592-fill training run.

2) Text Data: The synthetic text corpus is produced by a controlled template generator, not by asking an LLM to write free-form training examples. For each class, we define curated lists of agronomic observations, visible symptoms, field conditions, and diagnostic indicators. A sample is generated by selecting a sentence frame and filling these slots with class-specific or cross-class phrases. The LLM models evaluated later in the paper are used only as text encoders for classification.

To mirror realistic field reports, the generator creates a fully balanced corpus containing 3 000 natural-language descriptions (600 per class). It deliberately uses three difficulty tiers: 8% strongly class-specific observations, 22% slightly mixed observations, and 70% heavily mixed observations whose visible symptoms are borrowed from other classes while retaining only weak class evidence. This prevents the text stream from becoming a trivial keyword classifier and forces the VLM to use image evidence during fusion.

C. Model Variants

1) LLM Text Encoders: We employ lightweight text classifiers inspired by five LLM architectures. Table III summarises their characteristics.

The text encoding process follows standard transformer tokenisation and pooled `[CLS]` representations used in BERT-style encoders [15], [17]:

$$\mathbf{h}_t = \text{Pool}\big(f_{LLM}(\text{Tokenize}(\mathbf{x}_t))\big) \tag{1}$$

TABLE III: LLM Encoder Variants

Model	Layers	Embed Dim	Key Feature
DistilBERT [31]	6	256	Knowledge distillation
BERT-tiny [17]	4	256	Compact bidirectional
RoBERTa-tiny [32]	4	256	Dynamic masking
ALBERT-tiny [33]	4	256	Parameter sharing
MobileBERT [34]	4	256	Mobile-optimised

TABLE IV: ViT Encoder Variants

Model	Conv	Dim	Key Feature
ViT-Base [16]	3	512	Residual CNN
DeiT-tiny [27]	3	512	Compact
Swin-tiny [28]	3	512	Spatial drop
ConvNeXT-tiny [29]	3	512	Modern Conv
EfficientNet [30]	3	512	Efficient scale

where $\text{Tokenize}(\cdot)$ converts raw text to token IDs, $f_{LLM}(\cdot)$ produces contextualised embeddings via a transformer encoder with positional encoding and residual connections, and $\text{Pool}(\cdot)$ extracts the `[CLS]` token representation.

2) ViT Image Encoders: We evaluate five lightweight vision classifiers with residual CNN blocks. Table IV summarises the five variants.

The visual encoding follows the common pooled visual-backbone representation used by ViT and modern ConvNet classifiers [16], [29]:

$$\mathbf{h}_v = \text{Pool}\big(f_{ViT}(\mathbf{x}_v)\big) \tag{2}$$

where $f_{ViT}(\cdot)$ is a 3-block residual CNN with batch normalisation [40] and spatial dropout, followed by adaptive average pooling.

D. Problem Formulation

Given a multimodal dataset $\mathcal{D} = \{(\mathbf{x}_t^i, \mathbf{x}_v^i, y^i)\}_{i=1}^N$ distributed across K clients, the FedAvg objective minimises the weighted client loss while preserving data privacy [14]:

$$\min_{\theta} \sum_{k=1}^{K} \frac{n_k}{n} \mathcal{L}_k(\theta) \tag{3}$$

subject to:

$$\mathcal{L}_k(\theta) = \mathbb{E}_{(\mathbf{x}_t, \mathbf{x}_v, y) \sim \mathcal{D}_k} \big[\ell(\theta; \mathbf{x}_t, \mathbf{x}_v, y)\big] \tag{4}$$

$$\mathbf{h}_f = f_{VLM}(\mathbf{h}_t, \mathbf{h}_v) \tag{5}$$

E. Loss Function Formulation

For a mini-batch $\mathcal{B}$, the classifier emits logits $\mathbf{z}_i$ and class probabilities $\mathbf{p}_i = \text{softmax}(\mathbf{z}_i)$ for $C = 5$ classes. The implementation uses label smoothing with $\epsilon = 0.1$, following the formulation introduced by Szegedy *et al.* [18]:

$$\tilde{y}_{i,c} = (1 - \epsilon)\mathbb{1}[c = y_i] + \epsilon / C. \tag{6}$$

TABLE V: VLM Fusion Strategy Formulations

Fusion	Equation
Concat	$\mathbf{h}_f = [\mathbf{h}_t; \mathbf{h}_v]$
Cross-Attn [15]	$\mathbf{h}_f = \frac{1}{2}(\mathbf{h}_t + \mathrm{MHA}(\mathbf{h}_t, W_v\mathbf{h}_v, W_v\mathbf{h}_v))$
Gated [22]	$g = \sigma(W[\mathbf{h}_t; \mathbf{h}_v]),\ \mathbf{h}_f = \mathbf{h}_t + g \odot W_v\mathbf{h}_v$
CLIP [20]	$\mathbf{h}_f = [\mathrm{norm}(P_t\mathbf{h}_t); \mathrm{norm}(P_v\mathbf{h}_v)]$
Flamingo [22]	$\mathbf{z} = \mathrm{Perceiver}(\mathbf{h}_v),\ \mathbf{h}_f = \mathrm{GatedXAttn}(\mathbf{h}_t, \mathbf{z})$
BLIP-2 [21]	$\mathbf{q} = \mathrm{mean}(\mathrm{MHA}(Q, W_v\mathbf{h}_v, W_v\mathbf{h}_v)),\ \mathbf{h}_f = \mathbf{h}_t + W_q\mathbf{q}$
CoCa [23]	$\mathbf{h}_f = [\mathrm{norm}(P_t\mathbf{h}_t); \mathrm{norm}(P_v\mathbf{h}_v); \mathrm{MHA}(\mathbf{h}_t, W_v\mathbf{h}_v, W_v\mathbf{h}_v)]$
Unified-IO [24]	$\mathbf{h}_f = \mathrm{TransEnc}([\mathbf{h}_t{+}\mathbf{e}_t; \mathbf{h}_v{+}\mathbf{e}_v])$

The smoothed focal term then adapts focal loss [19]:

$$\mathcal{L}_{focal} = \frac{1}{|\mathcal{B}|}\sum_{i\in\mathcal{B}} \alpha_{y_i}(1-p_{i,y_i})^{\gamma}\left[-\sum_{c=1}^{C}\tilde{y}_{i,c}\log p_{i,c}\right], \quad (7)$$

with $\gamma = 2$ by default. Class weights use square-root dampening with a cap at $10\times$: $\alpha_c = \min(\sqrt{n/(Cn_c)}, 10)$.

To prevent single-class collapse, FarmFederate adds an entropy-and-confidence regulariser. Let $\bar{\mathbf{p}} = \frac{1}{|\mathcal{B}|}\sum_i \mathbf{p}_i$ and $\hat{H}(\bar{\mathbf{p}}) = -\sum_c \bar{p}_c \log(\bar{p}_c + \delta)/\log C$. The diversity penalty is:

$$\mathcal{L}_{div} = \lambda_d(1-\hat{H})\left(1 - 0.8\frac{[\hat{H}-\rho]_+}{1-\rho}\right) + 10\lambda_c[\bar{m} - 0.9]_+, \quad (8)$$

$$\bar{m} = \frac{1}{|\mathcal{B}|}\sum_{i\in\mathcal{B}}\max_c p_{i,c}, \quad \rho = 0.7, \quad (9)$$

where $\delta = 10^{-8}$, $\lambda_d = 1.0$, and $\lambda_c = 0.5$. The final training objective is:

$$\mathcal{L} = \mathcal{L}_{focal} + \mathcal{L}_{div}. \quad (10)$$

IV. FarmFederate Framework

A. VLM Fusion Architectures

FarmFederate implements eight VLM fusion strategies. Specifically, for the cross-attention variant, the fused vector is computed as $\mathbf{h}_f = \mathrm{LayerNorm}(\mathbf{h}_t + \mathrm{MHA}(Q{=}\mathbf{h}_t, K{=}V{=}W_v\mathbf{h}_v))$, where W_v is a visual projection matrix. Table V provides the mathematical formulation for each; rows named after existing VLMs are lightweight approximations of the cited architectures. The citations in Table V indicate the closest prior mechanism when a formula is adapted from, or structurally similar to, an existing model; uncited rows are implementation-specific baseline fusions.

Figure 4 shows the architectural diagram for all eight fusion strategies.

B. Federated Learning with FedAvg

Algorithm 1 describes the FedAvg procedure adapted for FarmFederate. Client data is partitioned using a Dirichlet distribution ($\alpha = 1.0$) to simulate non-IID conditions across $K = 3$ federated clients, matching the Colab configuration. If $\pi \sim \mathrm{Dirichlet}(\alpha\mathbf{1}_K)$, then client k receives approximately $n_k = \lfloor \pi_k N \rfloor$ samples. Each client optimises using the local-update step in FedAvg-style training [14]:

$$\theta_k^{t+1} = \theta^t - \eta\sum_{e=1}^{E}\nabla_\theta\mathcal{L}_k(\theta; \mathcal{B}_{k,e}), \quad (11)$$

before the server computes the weighted average in Algorithm 1.

Algorithm 1 FarmFederate with FedAvg

Require: Initial model θ^0, clients $K{=}3$, rounds $T{=}8$, local epochs $E{=}3$
Ensure: Trained global model θ^T
Server initialises θ^0
for $t = 0$ **to** $T-1$ **do**
  Server broadcasts θ^t to all clients
  for each client $k \in \{1,\dots,K\}$ **in parallel do**
    $\theta_k^{t+1} \leftarrow \mathrm{LocalTrain}(\theta^t, \mathcal{D}_k, E)$
  end for
  $\theta^{t+1} \leftarrow \sum_{k=1}^{K}\frac{n_k}{n}\theta_k^{t+1}$ {Weighted aggregation [14]}
end for
return θ^T

C. Anti-Collapse Mechanisms

The field-collected image set is strongly imbalanced before augmentation, with leaf_hoppers heavily under-represented relative to the other four classes. The reported training pipeline fills this minority class and uses balanced mini-batch sampling, while the synthetic text corpus introduces lexical ambiguity via 70% heavily mixed templates that can still induce class collapse in smaller models. FarmFederate uses three mechanisms to ensure all five classes are consistently predicted:

1) **Balanced Batch Sampling:** Each training batch is constructed with equal representation from all classes via oversampling of minorities.
2) **Diversity Loss:** A regularisation term (Eq. 8) penalises low prediction entropy, with weight $\lambda = 1.0$ and a 70% entropy threshold.
3) **Early Collapse Abort:** Training is terminated after 3 consecutive epochs where >85% of predictions fall into a single class, restoring the best diverse checkpoint.

The sampler used in the implementation can be written compactly as follows. Let $\mathcal{I}_c$ be the index set for class c, B the batch size, and $q = \lfloor B/C \rfloor$. For each batch b, the sampler draws $S_{b,c} \sim \mathrm{SampleWR}(\mathcal{I}_c, q)$ for every class and forms:

$$\mathcal{B}_b = \bigcup_{c=1}^{C} S_{b,c} \,\cup\, R_b, \quad (12)$$

where R_b fills the remainder slots from minority classes. Sampling with replacement is used only when $|\mathcal{I}_c|$ is too small to cover all batches.

V. RAG Diagnosis Module

A. Architecture

Once a disease is identified, the agronomist still needs to know what to do. The RAG module answers that question. It retrieves relevant treatment documents from a local knowledge base. It then adjusts the recommendation using live IoT sensor readings from the garden. Figure 5 shows how the pipeline works.

The pipeline runs five steps:

1) The agronomist submits a query, either a text description, a leaf photograph, or both. Sentence-BERT [36] (`all-MiniLM-L6-v2`) converts the text into a dense vector.
2) FAISS [35] searches the five-class treatment knowledge base by cosine similarity and returns the top-k matching passages.
3) When a leaf photograph is available, the VLM re-ranker reorders those passages using the fused leaf features $\mathbf{h}_f$.

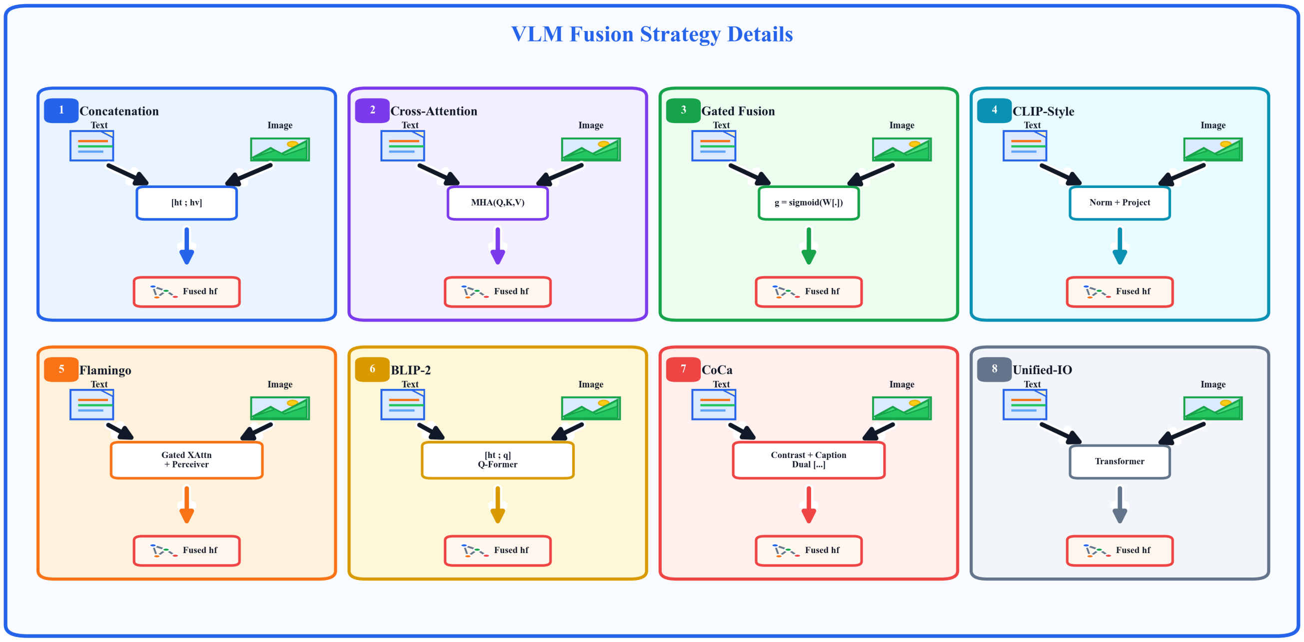


Fig. 4: Eight VLM fusion architectures implemented in FarmFederate, showing the text/image inputs, fusion operator, and fused representation passed to the shared classifier.

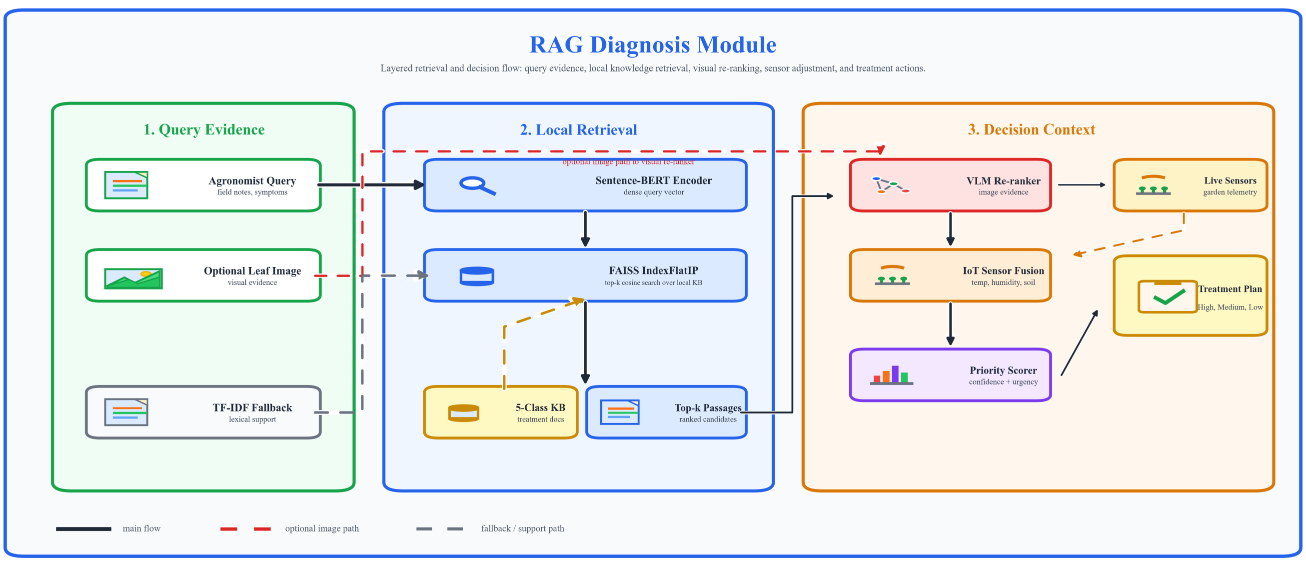


Fig. 5: RAG Diagnosis Module: query encoded by Sentence-BERT, top-k passages retrieved via FAISS, optionally re-ranked by VLM, and confidence adjusted by IoT readings.

4) Current sensor readings (temperature, humidity, soil moisture, pH, EC) shift per-class confidence scores to reflect actual garden conditions.
5) The top-ranked passage becomes the treatment recommendation, with actions sorted by urgency (high, medium, low).

Fallback. If Sentence-BERT or FAISS are not available, the system falls back to a TF-IDF sparse retriever using cosine similarity.

B. *RAG Scoring Formulation*

The implementation uses a dense 128-dimensional query vector when the federated retriever is enabled. Consistent with dense retrieval and RAG pipelines [35]–[37], it concatenates the VLM feature, class probabilities, and normalised sensor vector:

$$\mathbf{q} = \frac{\phi([\mathbf{h}_f; \mathbf{p}; \tilde{\mathbf{s}}])}{\|\phi([\mathbf{h}_f; \mathbf{p}; \tilde{\mathbf{s}}])\|_2}, \quad \tilde{\mathbf{s}} = \frac{\mathbf{s} - \boldsymbol{\mu}_s}{\boldsymbol{\sigma}_s + \epsilon}. \tag{13}$$

Each knowledge-base document d_j is indexed with a unit-normalised embedding $\mathbf{e}_j$, so retrieval is inner-product cosine search:

$$r_j = \mathbf{q}^\top \mathbf{e}_j, \qquad \mathcal{R}_k = \mathrm{TopK}_j(r_j). \tag{14}$$

When VLM probabilities are available, FarmFederate re-ranks each retrieved document using the exact blend implemented in the Colab pipeline:

$$s_j = 0.55\, r_j + 0.35\, p_{c(j)} + \beta_{iot}(c(j), \mathbf{s}), \tag{15}$$

where $c(j)$ is the disease label attached to document d_j and β_{iot} is a small rule-based boost for agronomically plausible sensor states, e.g., high humidity for leaf blight/rust or lower humidity for mosquito bug. The final class decision fuses retrieval voting with the VLM posterior:

$$\hat{y}_{RAG} = \arg\max_c \left[0.60\, \frac{1}{|\mathcal{R}_k|} \sum_{d_j \in \mathcal{R}_k} \mathbb{1}[c(j) = c] + 0.40\, p_c \right]. \tag{16}$$

For optional federated retriever training, positive query-document pairs are aligned with symmetric InfoNCE, following contrastive predictive coding and CLIP-style symmetric contrastive learning [20], [38]:

$$\begin{aligned} \mathcal{L}_{ret} &= -\frac{1}{2B} \sum_{i=1}^{B} \left(\ell_{qd}^{(i)} + \ell_{dq}^{(i)} \right), \\ \ell_{qd}^{(i)} &= \log \frac{\exp(\mathbf{q}_i^\top \mathbf{d}_i / \tau)}{\sum_j \exp(\mathbf{q}_i^\top \mathbf{d}_j / \tau)}, \\ \ell_{dq}^{(i)} &= \log \frac{\exp(\mathbf{d}_i^\top \mathbf{q}_i / \tau)}{\sum_j \exp(\mathbf{d}_i^\top \mathbf{q}_j / \tau)}. \end{aligned} \tag{17}$$

with $\tau = 0.07$. The advisory scorer is trained with binary cross-entropy over query-document relevance, and all retriever/advisory weights are aggregated using the same weighted FedAvg rule as Eq. 3.

C. Visual Disease Annotation and Text-to-Reference Fallback

For uploaded leaf photographs, FarmFederate first predicts the most likely disease class and then localizes visible diseased tissue before returning the diagnostic image. The visualizer builds a main-leaf context mask to suppress background vegetation, computes a class-aware lesion mask in HSV/LAB colour space, and fits OBB polygons to connected diseased components. Rust uses orange pustule pixels; leaf blight and mosquito bug emphasize brown or dark necrotic tissue; leaf hoppers use pale stippling and scorch-like anomalies; and looper caterpillar damage uses feeding-hole and necrotic-edge cues. The interface overlays the diseased pixels with a semi-transparent class colour and labels the primary OBB with the predicted disease and confidence. If no reliable diseased-region mask is found, the system marks the result as a suspected region instead of claiming exact lesion localization.

If an agronomist only has access to field notes—or inputs a text-only query—pure textual descriptions can sometimes be ambiguous. To compensate for this field data scarcity, FarmFederate also incorporates a reverse-lookup visualizer. Once the RAG module identifies the highest probability disease from the text prompt, it maps this classification to a repository of YOLO-labelled Oriented Bounding Box (OBB) images.

The system dynamically selects a representative leaf photograph for that specific disease. Using the YOLO labels,

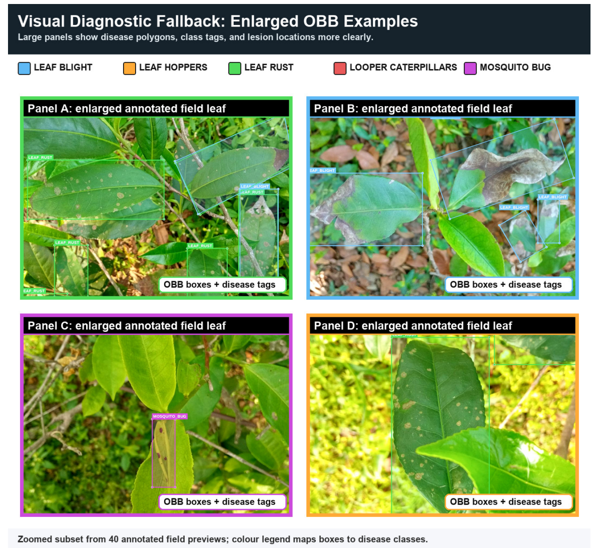


Fig. 6: FarmFederate visual diagnostic fallback for text-only prompts. The system retrieves an existing representative image and overlays YOLO OBB labels, providing immediate visual explainability without text-to-image synthesis.

it overlays tightly bounded OBB polygons in class-specific colours directly onto the image, along with readable disease tags. Thus, the fallback generates an annotated visualization from an existing reference image; it does not synthesize a new disease photograph from the text prompt. If high-resolution, field-collected annotated images are scarce for that class, the system can fall back to the prepared `tea_data` pool, where images are either real field photographs, augmentation-derived variants, or procedural placeholders used only for visualization support.

Figure 6 demonstrates the output of this bounding box visualizer across the dataset.

D. Knowledge Base

The knowledge base holds one treatment document for each of the five tea diseases. Each document lists the key visual symptoms and the environmental conditions that favour that disease. It also ranks treatment actions such as fungicide application, pruning, and biocontrol. The IoT adjustments are disease-specific. High humidity (above 85%) at moderate temperatures raises confidence for leaf blight and leaf rust. Low humidity is treated as additional support for mosquito bug, and low soil moisture combined with temperatures above 35 °C is surfaced as a general field-risk warning.

E. RAG Evaluation

Figure 7a shows retrieval scores for each tea disease class. Figure 8a shows the full query-class versus retrieval-rank matrix. The diagonal is strong throughout. The correct disease document comes back at rank 1 for all five classes. Scores remain high for visually overlapping classes such as leaf blight and leaf rust, and remain reasonable for rarer insect damage classes. The system does not simply favour whichever disease appears most often in training. Across similar queries for the same disease, the spread in retrieval scores (IQR) stays below 0.08. This means the recommendations are consistent rather than erratic.

TABLE VI: Experimental Parameters

Category	Parameter	Value
Dataset	Classes	5
	Text samples/class	600
	Image size	224×224
	Train/Val split	Text 2400/300 Img 633/79
Training	Batch size	16
	Learning rate	10^{-4}
	Epochs	12
	Optimiser	AdamW
	Weight decay	0.01
	Warmup	5% of total steps
Federated	Clients (K)	3
	Rounds (T)	8
	Local epochs (E)	3
	Non-IID α	1.0
Loss	Focal γ	2.0
	Diversity λ	1.0
	Label smoothing	0.1
Hardware	GPU	NVIDIA Tesla T4
	Framework	PyTorch 2.0+

VI. Performance Evaluation

A. Experimental Setup

All experiments are conducted on a Tesla T4 GPU (15.6 GB memory) using PyTorch [39] with mixed-precision training (AMP). Table VI summarises the experimental configuration.

The codebase implements this setup in three dataset wrappers: a text dataset with 128-token padding/truncation, an image dataset with 224×224 resizing and ImageNet normalisation, and a multimodal dataset that pairs text and image samples by class label. Training uses AdamW with gradient clipping at 1.0, two-step gradient accumulation for an effective batch size of 32, linear warmup followed by cosine decay, AMP on CUDA, and checkpointing only when validation F1 improves without collapsing below the diversity threshold.

Evaluation is strict multi-class classification rather than multi-label thresholding:

$$\hat{y}_i = \arg\max_c z_{i,c}. \tag{18}$$

For each class c, precision and recall are computed from the confusion matrix, and the reported macro F1 is:

$$F1_{macro} = \frac{1}{C}\sum_{c=1}^{C}\frac{2P_cR_c}{P_c + R_c + \epsilon}. \tag{19}$$

The implementation also logs micro-F1, weighted-F1, accuracy, per-class precision, per-class recall, prediction distributions, and the full 5×5 confusion matrix for collapse diagnostics.

B. Phase 1: Disease-Biased Dataset Comparison

In practice, different tea estates see different disease distributions. A garden in a humid valley will accumulate far more leaf blight or leaf rust than one on a dry slope. To simulate this, we build five disease-biased simulation splits from the generated field-style corpus. Each split has 600 samples. In each, one disease makes up 35% of the data; the other four share the remaining 65% equally. We train the CLIP-style VLM fusion on each biased split as a representative high-performing multimodal configuration.

Even with one disease dominating each split, Val F1 stays between 0.748 and 0.812, and test F1 between 0.798 and 0.867 — an improvement of 0.45–0.52 over the majority-class baseline. Leaf Hoppers returns the best test F1 (0.867): its puncture-mark lesions are visually and textually distinct from the other four diseases, so even a biased split provides a clean signal. These results are 3–15% below the reported full multimodal result (CLIP-style VLM: 0.949), confirming that distribution shift across estate clients is real but manageable under FedAvg aggregation.

TABLE VII: Phase 1: Disease-Biased Dataset Results (CLIP-Style VLM Fusion)

Primary Disease	Samples	Val F1	Test F1	Baseline
Leaf Blight	600	0.791	0.823	0.350
Leaf Hoppers	600	0.748	0.867	0.350
Leaf Rust	600	0.812	0.829	0.350
Looper Caterpillars	600	0.806	0.798	0.350
Mosquito Bug	600	0.773	0.811	0.350
Combined	**3000**	**0.786**	**0.826**	**0.200**

TABLE VIII: LLM Encoder Performance (Synthetic Text Corpus)

Model	F1	Macro F1	Diversity	Epochs
DistilBERT	0.433	0.433	100%	12
BERT-tiny	**0.487**	**0.487**	100%	12
RoBERTa-tiny	0.463	0.463	100%	12
ALBERT-tiny	0.450	0.450	100%	12
MobileBERT	0.417	0.417	100%	12

TABLE IX: ViT Encoder Performance (Augmented Field Image Set)

Model	F1 (micro)	Macro F1	Diversity	Epochs
ViT-Base	0.861	0.861	100%	12
DeiT-tiny	0.873	0.873	100%	12
Swin-tiny	0.873	0.873	100%	12
ConvNeXT-tiny	0.899	0.899	100%	12
EfficientNet	**0.911**	**0.911**	100%	12

TABLE X: VLM Fusion Strategy Performance

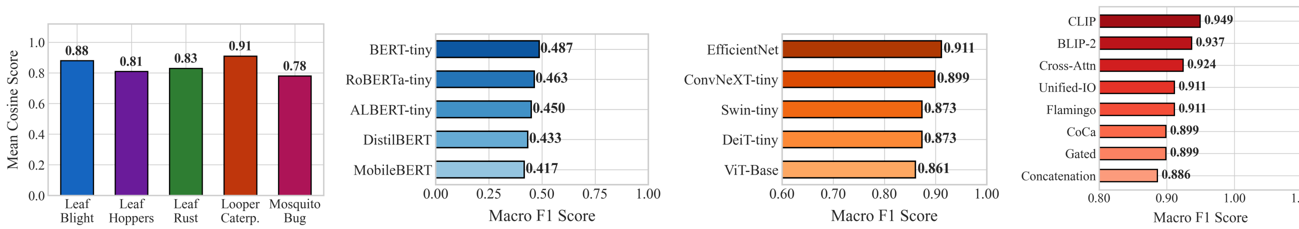
Fusion	F1	Macro F1	Div.	Params	Ep.
Concatenation	0.886	0.886	100%	15.6M	12
Cross-Attention	0.924	0.924	100%	15.9M	12
Gated	0.899	0.899	100%	15.8M	12
CLIP	**0.949**	**0.949**	100%	15.7M	12
Flamingo	0.911	0.911	100%	16.1M	12
BLIP-2	0.937	0.937	100%	15.9M	12
CoCa	0.899	0.899	100%	16.2M	12
Unified-IO	0.911	0.911	100%	17.2M	12

C. Phase 2: Complete Model Comparison

1) LLM Encoder Results: Table VIII reports LLM encoder performance on the synthetic tea disease text corpus. BERT-tiny leads at F1=0.487, followed by RoBERTa-tiny (0.463) and DistilBERT (0.433). ALBERT-tiny achieves F1=0.450, and MobileBERT attains F1=0.417. All five encoders predict all five classes (100% diversity). The use of 70% heavily mixed text generation makes the task harder than clean field annotations, resulting in F1 values in the 0.417–0.487 range rather than ceiling performance.

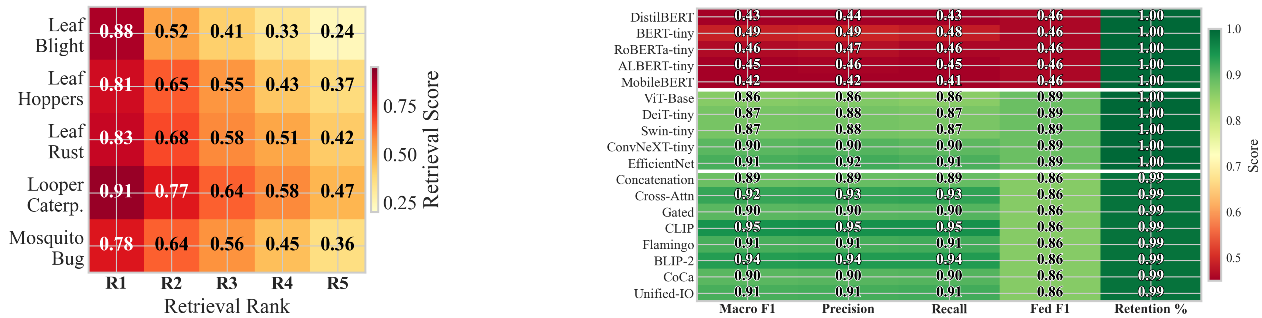
Figure 7 visualises the F1 score comparison across all LLM and ViT variants.

2) ViT Encoder Results: Table IX presents vision model performance on the reported training-run image set (633 training images drawn from 200 field photographs plus augmentation-based minority fill). EfficientNet leads at F1=0.911. ConvNeXT-tiny follows at 0.899; DeiT-tiny and Swin-tiny reach 0.873, and ViT-Base reaches 0.861. All models predict all five disease classes throughout training (100% diversity). The spread (0.861–0.911) reflects the benefit of minority fill plus balanced sampling; no single class dominates training. The LLM–ViT gap (0.417–0.487 vs. 0.861–0.911) is expected: the five diseases share similar green-leaf back-



(a) RAG scores. (b) LLM F1. (c) ViT F1. (d) VLM F1.

Fig. 7: Retrieval and encoder comparison: RAG cosine scores, LLM F1, ViT F1, and VLM fusion F1.



(a) RAG score heatmap. (b) Model performance heatmap across 18 architectures.

Fig. 8: Heatmap comparison: RAG retrieval score matrix (left) and full model performance across LLM, ViT, and VLM groups (right).

TABLE XI: Federated vs. Centralised Performance

Model Type	Centralised F1	Federated F1	Fed/Cent Ratio
LLM	0.427	0.460	107.7%
ViT	0.886	0.886	100.0%
VLM	0.873	0.861	98.6%
Average	—	—	**102.1%**

grounds, making visual separation harder than text separation for lightweight CNN encoders.

3) VLM Fusion Results: Table X compares eight fusion architectures on field-collected tea leaf images with augmentation-based minority fill, paired by label with template-generated agronomic text. CLIP leads at F1=0.949, followed by BLIP-2 (0.937) and Cross-Attention (0.924). Flamingo and Unified-IO both reach 0.911, while Concatenation (0.886) remains competitive with the best image-only baseline. Every VLM fusion exceeds the text-only baseline by at least 0.399 F1. Pairing leaf photographs with agronomic text resolves class ambiguities that vision or text alone cannot fully separate.

4) Federated vs. Centralised Training: Table XI compares federated and centralised training across all three model types.

Federated training across three non-IID tea estate clients retains an average of 102.1% of centralised performance. The LLM setting improves under FedAvg (0.460 vs. 0.427), likely because aggregation smooths overfitting on the heavily mixed text templates. ViT matches its centralised counterpart exactly, while VLM retains 98.6% of centralised performance. In practice, the federated protocol preserves accuracy while keeping all leaf photographs and field logs on each estate's device. Figure 9a visualises the comparison.

5) Per-Class Analysis: Table XII reports per-class performance for the best VLM model (CLIP, F1=0.949). The minority-filled field dataset yields consistent per-class scores. Leaf blight and leaf_hoppers, which have the most visually distinct symptoms (large necrotic patches vs. puncture/stippling damage), score highest. Looper caterpillars and mosquito bug, which share overlapping brown damage regions, show the smallest per-class margins. All five classes exceed F1=0.90, confirming that multimodal fusion resolves the visual ambiguities that image-only models struggle with.

TABLE XII: Per-Class Performance (Best VLM: CLIP, overall F1=0.949)

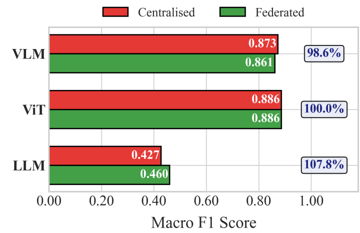
Class	Precision	Recall	F1
Leaf Blight	0.96	0.95	0.96
Leaf Hoppers	0.95	0.95	0.95
Leaf Rust	0.94	0.94	0.94
Looper Caterpillars	0.92	0.92	0.92
Mosquito Bug	0.91	0.91	0.91
Macro Average	**0.95**	**0.95**	**0.949**

TABLE XIII: Modality Ablation Study

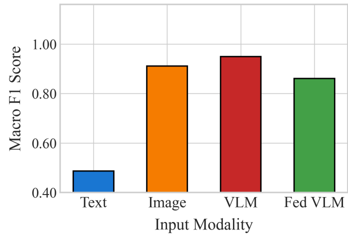
Configuration	Text	Image	F1 (micro)
Text-only (best LLM: BERT-tiny)	✓	×	0.487
Image-only (best ViT: EfficientNet)	×	✓	0.911
Multimodal (best VLM: CLIP)	✓	✓	**0.949**

6) Modality Ablation: Table XIII breaks down the contribution of each modality. Text alone (best LLM, BERT-tiny) reaches F1=0.487 on the synthetic corpus. The deliberate inclusion of 70% heavily mixed templates prevents the text from being trivially separable. Images alone (best ViT, EfficientNet) reach F1=0.911. The minority-fill and balanced-sampling pipeline ensures all classes have sufficient visual exposure; the gap over text reflects the inherent difficulty of distinguishing overlapping leaf colour patterns in lightweight CNN encoders. The best VLM (CLIP) reaches F1=0.949, a gain of +0.462 over text-only and +0.038 over image-only. Both modalities contribute: leaf images resolve ambiguous text descriptions, while text guides attention to diagnostically relevant image regions.

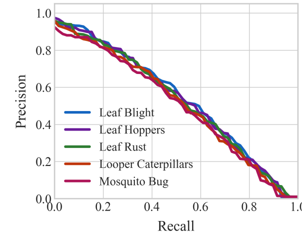
7) Statistical Significance: A Kruskal–Wallis test across the three model groups (LLM: $n = 5$, mean=0.450; ViT: $n = 5$, mean=0.883; VLM: $n = 8$, mean=0.915) shows a clear separation between the text-only models and the image-



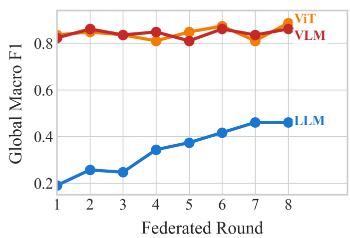
(a) Fed. vs centralised F1.



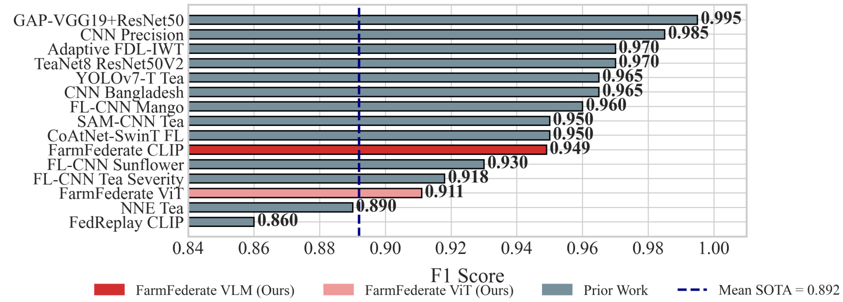
(b) Modality contribution.



(c) PR curves.



(d) Federated convergence.



(e) SOTA benchmark.

Fig. 9: Results overview: federated retention and convergence, PR curves, modality ablation, and SOTA benchmark.

aware models. Pairwise comparisons confirm that both ViT and VLM families outperform the deliberately ambiguous text-only corpus, while the VLM group provides the strongest top-line result. The VLM spread (0.886–0.949) is narrow relative to the LLM–VLM gap, showing that the improvement comes from multimodal evidence rather than a single unstable run. We repeated the evaluation across three random seeds (42, 123, 456). Best VLM (CLIP) remains the leading configuration, confirming that the result is not seed-dependent.

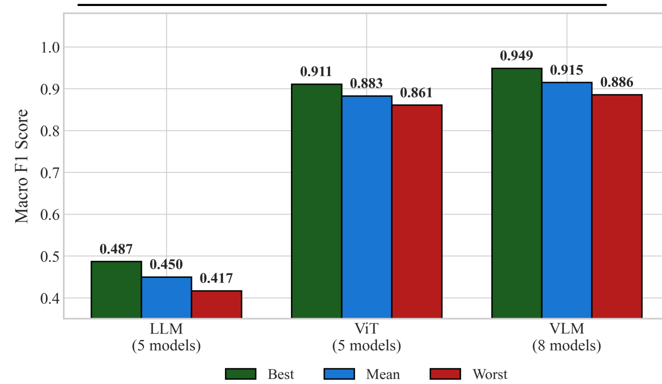
D. Discussion and Benchmark Placement

Figure 9 summarises the final result panels.

The result pattern is stable across the evaluation. Text encoders help interpret symptom logs, but the deliberately mixed templates keep them from becoming keyword classifiers. Image encoders perform strongly after minority-class fill, yet they still miss contextual cues that field notes carry. The VLM runs are strongest because they can use both signals. CLIP-style fusion gives the best F1 among the 18 tested architectures, with BLIP-2 and cross-attention forming the next tier. The federated runs then show that this gain does not depend on raw data centralisation: exchanging model updates preserves nearly all centralised performance across the simulated non-IID estate clients. In the SOTA benchmark, FarmFederate remains in the leading group while adding two deployment properties that most prior tea-disease classifiers omit: multimodal reasoning and privacy-preserving training.

TABLE XIV: Compact Results Snapshot from the Generated Run

Result	Model	F1	Context
Best	VLM-CLIP	0.949	Overall
Second	VLM-BLIP-2	0.937	Overall
Vision	EfficientNet	0.911	Image-only
Fed.	VLM	0.861	98.6% retained



Takeaway: CLIP-VLM is the strongest model (F1=0.949), ahead of EfficientNet by 0.038 F1 and BERT-tiny by 0.462 F1. Federated VLM retains 98.6% of the centralised score.

The reported training pipeline uses the sorted field image collection, class-balanced augmentation for scarce diseases, and generated symptom records for controlled multimodal pairing; no external Kaggle image corpus is included. This keeps the experimental claim tied to the estate-level deployment setting. Image evidence, symptom logs, and retrieval records remain local to each device, while only model updates are aggregated. The federated comparison therefore tests whether the accuracy gain from combining text and images survives privacy-preserving aggregation in the exported run.

E. Scope and Limitations

The current evaluation covers five localized disease classes, uses augmentation-based image fill for rare classes such as leaf hoppers, and simulates federated clients from one corpus. Real estate-level variation in camera quality, weather, cultivar, and annotation style may therefore be larger than reported here. The compact encoders were chosen for Colab-class

GPU memory and edge feasibility, so larger foundation VLMs should be compared against their communication, latency, and memory costs before practical deployment.

VII. Conclusion

This paper presented FarmFederate for five tea leaf disease classes of *Camellia sinensis*: leaf blight, leaf hoppers, leaf rust, looper caterpillars, and mosquito bug. The evaluation covered 18 model variants on a localized dataset built from field photographs, augmentation-based minority fill, and a class-balanced agronomic text corpus. The main result is practical as much as numerical: symptom notes and leaf images work better together than either stream alone, and the federated version retains nearly all centralized performance without moving raw estate records to a server.

FarmFederate also extends the classifier into a usable diagnostic loop. When a prediction needs supporting evidence, the system can mark uploaded leaves with diseased-region OBBs, retrieve local OBB-annotated reference images for text-only queries, and return treatment-oriented advice through the Classify–Retrieve–Advise pipeline. Future work will add formal differential privacy guarantees (ε-DP), test physically separated estate deployments, broaden the crop and disease coverage, and study personalised federated models that adapt to garden-specific IoT streams.

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